

# NAGAPPAN S

## Full Stack Engineer | AI Engineering

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### 👤 PROFESSIONAL SUMMARY

Seasoned Full Stack Engineer with 9+ years of experience building scalable web applications across fintech, insurance, and enterprise domains. Proficient in React, Node.js, Next.js, and cloud technologies, with a strong track record of leading teams and delivering complex migrations end to end. Recently expanded into AI Engineering building RAG pipelines, agentic workflows, and LLM-powered applications using Anthropic Claude, LangChain, and pgvector. Actively seeking senior engineering roles where I can bridge full stack expertise with modern AI capabilities to deliver intelligent, high-impact products.

### 📁 WORK EXPERIENCE

01/2025 – Present

#### **Capgemini Technology Services India Limited** **Senior Consultant**

- Leading offshore development of Synchrony's Shopify Marketing and Payment applications, enabling merchants to seamlessly integrate financing options into their e-commerce storefronts.
- Architected and implemented an iFrame-based Common UI system adopted across multiple Shopify apps, standardising configuration workflows, widget placements, and across merchant platforms.
- Designed and optimised backend APIs handling high-volume payment and marketing flows, improving reliability and reducing error rates across critical transaction pipelines.
- Drove application performance improvements by migrating legacy class-based React components to modern functional patterns, significantly improving maintainability and test coverage.
- Represented the team in internal Hackathons and Coding Quests, delivering innovative proof-of-concept solutions recognised at the project level.

06/2023 – 12/2024

#### **Cognizant Technology Solutions** **Senior Software Engineer**

- Led a 4-member offshore team as technical lead, responsible for delivery, code quality, and junior developer mentorship across a large-scale enterprise application.
- Spearheaded the end-to-end migration of a complex React application to Liberty Mutual's proprietary design system (LMDS), ensuring zero functional regression across all migrated modules.
- Drove a full architectural migration from legacy React patterns to Next.js 14 App Router, modernising the frontend stack and improving performance and SEO capabilities.
- Engineered an automated Service Desk ticket creation system integrated with JIRA, reducing manual effort and improving incident response time for operations teams.

- Implemented automated customer email notification services, improving communication workflows triggered by key application events.

10/2020 – 05/2023

**Saksoft Limited**  
**Technical Consultant**

- Built efficient, reusable, and accessible frontend systems using React, delivering consistent and performant user experiences across multiple client projects.
- Developed and secured REST APIs using Node.js with JWT-based authentication, ensuring protected access to sensitive application data.
- Integrated frontend applications with relational databases, implementing optimised queries for reliable data retrieval and storage.
- Applied performance optimisation techniques including lazy loading, code splitting, and memoisation, measurably improving application load times and user experience.

12/2017 – 10/2020

**Fujitsu Consulting India Private Limited**  
**Assistant Application Developer**

- Developed dynamic, cross-browser compatible user interfaces using HTML5, CSS3, JavaScript, and Bootstrap.
- Built interactive UI components leveraging jQuery, enhancing usability across client-facing dashboards.
- Delivered a full-stack proof-of-concept application using Node.js, implementing MVC architecture, template engines, and Express validations end to end.

03/2017 – 09/2017

**Tech Mahindra**  
**Associate Software Engineer**

- Provided production support and bug resolution for a live enterprise application, ensuring stability and minimal downtime.
- Delivered targeted feature enhancements on existing modules with minimal disruption to the production environment.

## SKILLS

### Frontend

HTML5, CSS3, JavaScript, React, Next.js, Tailwind, CSS, jQuery

### Databases

PostgreSQL, MongoDB, MySQL, pgvector

### Backend

Node.js, Express.js, REST APIs, JWT Authentication

### Cloud & DevOps

AWS (Certified Developer Associate), Docker, GitHub Actions (CI/CD)

### AI & LLM Engineering

LangChain, LangGraph, RAG Pipelines, Vector Search (pgvector), Prompt Engineering, Anthropic Claude API, Voyage AI Embeddings, Agentic AI, Multi-Agent Systems, Vercel AI SDK

### Tools & Platforms

Prisma ORM, NextAuth.js, Shopify, Git

## CERTIFICATIONS

### AWS Certified Developer Associate

(Jun 2020 - Jun 2023, Renewal planned)

## **Independent AI Engineering Projects**

[github.com/Nagappan03](https://github.com/Nagappan03)

### **RAG-Based Internal Document Q&A**

- Built a full-stack document intelligence app where users upload PDFs and query them via a chat interface. Stack: Next.js 14, Anthropic Claude Haiku, Voyage AI embeddings, pgvector on PostgreSQL, LangChain, Prisma, NextAuth.js. Implemented the complete RAG pipeline including chunking, embedding, vector similarity search, and streamed LLM responses.

### **AI-Powered SQL Query Generator**

- Natural language to SQL agent connected to a live PostgreSQL database. Users describe what they need in plain English; the agent generates the query using Claude's tool calling, executes it, and returns results with an explanation. Demonstrates function calling and agentic tool use patterns.

### **Multi-Agent Research Assistant**

- Autonomous research agent built with LangGraph. Given a topic, multiple specialised agents collaborate to search the web, synthesise findings, and produce a structured report. Demonstrates multi-agent orchestration, state management, and role-based agent design.

### **AI Job Application Copilot**

- LLM-powered tool that takes a job description and resume as inputs and generates tailored resume bullet points, a custom cover letter, and role-specific interview prep questions. Demonstrates multi-step LLM chaining and prompt engineering patterns.

## **Client Projects**

### **Synchrony eCarts - Shopify**

**(Jan 2025 - Present)**

- Designed and implemented an iFrame-based Common UI for consistent configuration management across both Marketing & Payment UI apps.
- Developed and optimised backend APIs for widget placements and payment configurations.
- Refactored legacy components and improved application performance and reliability.
- Collaborated with Synchrony stakeholders to drive enhancements, new feature rollouts, and cross-platform alignment.

### **Electronic Records Management (ERM) - Cleanup Crew**

**(Jul 2023 - Dec 2024)**

- Migrated the entire Admin UI from Vanilla React to Liberty Mutual's design system (LMDS).
- Built a module to automate JIRA Service Desk ticket creation and reflect ticket status in the UI.
- Led migration of the application from legacy React patterns to Next.js 14 App Router architecture.
- Delivered various visual and functional enhancements across the front-end application.

### **TransUnion CIBIL - ATLAS**

**(Sep 2021 - May 2023)**

- Implemented various async patterns (Promise chaining, async/await, Axios, Fetch API) for external API integrations.
- Improved application performance using lazy loading, code splitting, and memoisation techniques.
- Implemented Redux and Redux-Thunk for complex state management across the application.

### **Workruit - Job Search Platform**

**(Mar 2020 - Sep 2021)**

- Implemented secure user authentication using JWT and bcrypt.
- Built custom middleware ensuring only authenticated and authorised users could access protected routes.
- Developed and fine-tuned search algorithms matching user queries with job postings.